Multi-Interval Report

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Abstract

Through our circuit and population model, we want to show

1. The practical benefits of allowing for two and three free intervals compared to single intervals.
2. The trade off of each interval choices. This is done by an attempt to show the space of objective values by possible parameters.

1 Model

Circuit and population. A circuit has N customers of class $j \in \{S, M, L\}$ with current I_j and load $P_j = VI_j$. Its composition $\boldsymbol{n} = (n_S, n_M, n_L)$, $\sum_j n_j = N$, is random:

$$\boldsymbol{n} \sim \text{Trinomial}(N; p_S, p_M, p_L), \qquad w(\boldsymbol{n}) = \frac{N!}{n_S!\, n_M!\, n_L!}\, p_S^{n_S} p_M^{n_M} p_L^{n_L}. \tag{1}$$

Intervals. A design has $K \in \{1, 2, 3\}$ free intervals $\mathcal{I}_k = [s_k, e_k)$ of 3 hours. Each medium and large customer is assigned one interval; inside it the customer draws I_j, outside it I_S. An allocation $\boldsymbol{a} = (m_1, \ldots, m_K, l_1, \ldots, l_K)$ gives the medium and large customers in each interval, with $\sum_k m_k = n_M$ and $\sum_k l_k = n_L$. The current the interval customers add at time t, the circuit current and its peak are

$$\Delta A_{\boldsymbol{a}}(t) = \sum_{k:\, t \in \mathcal{I}_k} \big((I_M - I_S)\, m_k + (I_L - I_S)\, l_k\big), \tag{2}$$

$$A_{\boldsymbol{a}}(t) = NI_S + \Delta A_{\boldsymbol{a}}(t), \qquad \hat{A}(\boldsymbol{a}) = \max_t A_{\boldsymbol{a}}(t). \tag{3}$$

2 Objectives

Probability of overload. Each circuit composition $\boldsymbol{n}$ uses an allocation $\boldsymbol{a}(\boldsymbol{n})$. It overloads if its peak exceeds the limit T:

$$r(\boldsymbol{n}) = \begin{cases} 1, & \hat{A}(\boldsymbol{a}(\boldsymbol{n})) > T, \\ 0, & \text{otherwise}, \end{cases} \qquad O_{\text{viol}} = \sum_{\boldsymbol{n}} w(\boldsymbol{n})\, r(\boldsymbol{n}). \tag{4}$$

Solar soak. A share u of Victoria's N_{VIC}/N circuits is on the plan (uptake). The plan load is the total load of these circuits, every customer's I_S plus the current the intervals add,

$$L(t) = u\, \frac{N_{\text{VIC}}}{N} \sum_{\boldsymbol{n}} w(\boldsymbol{n})\, VA_{\boldsymbol{a}(\boldsymbol{n})}(t). \tag{5}$$

Over the hours the design's intervals cover, $\mathcal{U} = \bigcup_k \mathcal{I}_k$, the net soak is the rooftop solar the plan load soaks, less the excess load E that rooftop solar does not cover,

$$O_{\text{soak}} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \sum_{t \in \mathcal{U}} \min\big(L(t), X_d(t)\big) \Delta t - E, \tag{6}$$

$$E = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \sum_{t \in \mathcal{U}} \max\big(L(t) - X_d(t),\, 0\big) \Delta t, \tag{7}$$

where $X_d(t)$ is the Victorian rooftop solar generation on day d and $\mathcal{D}$ are the days of the year. The grid must supply the excess load. Results give O_{soak} as a percentage of $O^{\star}_{\text{soak}} = 14{,}807$ MWh/day, the mean rooftop solar between 08:00 and 16:00, which is the most any design could soak.

3 Experimental Results

3.1 Setup

3.1.1 Parameters

Parameter	Value	Note
N	40	customers per circuit
T	1000 A	circuit limit
(I_S, I_M, I_L)	(20, 40, 100) A	current per customer class
V	240 V	
(p_S, p_M, p_L)	(0.69, 0.30, 0.01)	today's mix
N_{VIC}	2.9×10^6	Victorian customers (72,500 circuits)
u	12.8%	uptake: 9,271 circuits on the plan, calibrated (Section 3.1.3)
$X_d(t)$	2025 VIC1 rooftop solar	Open Electricity, 5-minute steps, UTC+10
Δt	5 min	rooftop values change every 30 minutes
Usage	All-On	full current for the whole interval

3.1.2 Interval choice

Each season s (summer: Dec–Feb, autumn: Mar–May, winter: Jun–Aug, spring: Sep–Nov) has its own intervals. They are chosen to cover the most rooftop solar: with $\bar{X}_s(t)$ the mean rooftop solar of season s at time t, the chosen set maximises

$$S_s(\mathcal{I}_1, \ldots, \mathcal{I}_K) = \sum_{k=1}^{K} \sum_{t \in \mathcal{I}_k} \bar{X}_s(t)\, \Delta t, \tag{8}$$

over 3-hour intervals that start every 0.5 hours and lie inside 08:00–16:00 (ties: the most rooftop solar covered by their union). Two intervals do not overlap; of three intervals, the first and last do not overlap and the middle one overlaps both. With one set per season, a circuit overloads if it overloads in any season, and soak uses each season's intervals on that season's days.

Season	1 interval	2 intervals	3 intervals
Summer	11:30–14:30	10:00–13:00, 13:00–16:00	10:00–13:00, 11:30–14:30, 13:00–16:00
Autumn	11:30–14:30	10:00–13:00, 13:00–16:00	10:00–13:00, 11:30–14:30, 13:00–16:00
Winter	11:00–14:00	10:00–13:00, 13:00–16:00	10:00–13:00, 11:00–14:00, 13:00–16:00
Spring	11:00–14:00	09:30–12:30, 12:30–15:30	09:30–12:30, 11:00–14:00, 12:30–15:30

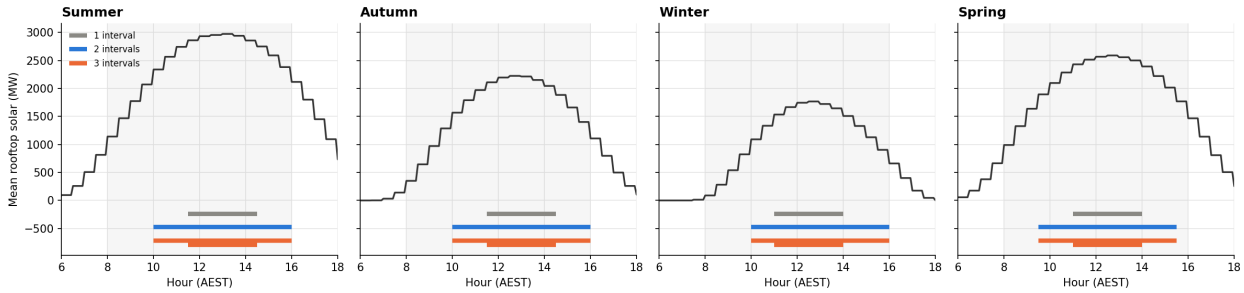


Figure 1: Mean daily rooftop solar per season. Bars: the chosen intervals (3 intervals: the middle interval is drawn lower); shaded: 08:00–16:00.

3.1.3 Seasonal scaling

The plan load $L(t)$ is the same in every season, but rooftop solar is not: on average 24.4 GWh/day in summer, 19.7 in spring, 14.7 in autumn and 10.2 in winter (Figure 1). Rooftop output exceeds the plan load for most of the covered hours in summer and spring, about matches it at midday in autumn and stays below it all day in winter (Figure 2), so solar soak is measured unevenly across seasons: without scaling, rooftop solar covers 95% of the plan load in the covered hours in summer, 90% in spring, 83% in autumn and 67% in winter (two intervals, today's mix).

We therefore scale the load counted for soak in each season s by the season's midday peak $\hat{X}_s = \max_t \bar{X}_s(t)$, the peak of its mean daily rooftop output,

$$L_s(t) = w_s\, L(t), \qquad w_s = \frac{\hat{X}_s}{\hat{X}}, \tag{9}$$

$$O_{\text{soak}} = \frac{1}{|\mathcal{D}|} \sum_s \sum_{d \in \mathcal{D}_s} \sum_{t \in \mathcal{U}_s} \Big[\min\big(L_s(t), X_d(t)\big) - \max\big(L_s(t) - X_d(t),\, 0\big) \Big] \Delta t, \tag{10}$$

where $\hat{X}$ is the mean of $\hat{X}_s$ over the days of the year, $\mathcal{D}_s$ the days of season s and $\mathcal{U}_s$ the hours its intervals cover. This gives $w = (1.25, 0.93, 0.74, 1.09)$ for summer, autumn, winter and spring, averaging 1 over the year. Every season then has the same ratio of plan load to midday rooftop peak. The scaling changes only the load counted for soak; circuit currents and overload are unchanged. Figures 2 and 3 show the plan load before and after the scaling (two intervals, today's mix), drawn over the hours where soak is counted. With the scaling rooftop solar covers 91%, 86%, 81% and 87% of the plan load in the covered hours in summer, autumn, winter and spring.

Uptake. The uptake is calibrated to the solar the plan should absorb: under the policy (one interval, every medium and large customer in it) at today's mix, the scaled plan load peaks at each season's midday peak, 2,974 MW in summer, 2,225 MW in autumn, 1,768 MW in winter and 2,589 MW in spring. This gives $u = 12.8\%$ (9,271 circuits, 371,000 customers). The load of every customer at I_S is then 2,219 MW in summer, below the mean rooftop output in the middle of the day in every season, so the free intervals have solar left to soak. The same uptake is used for every design, so two and three intervals spread the same customers over more hours.

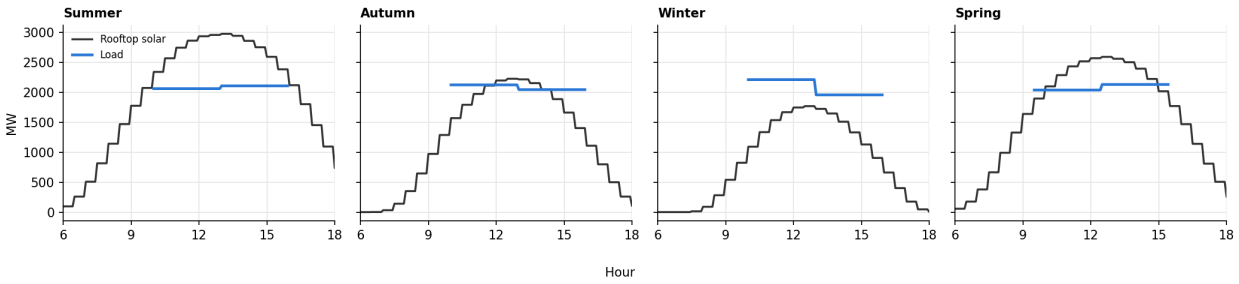


Figure 2: Before the scaling: mean rooftop solar and the plan load L.

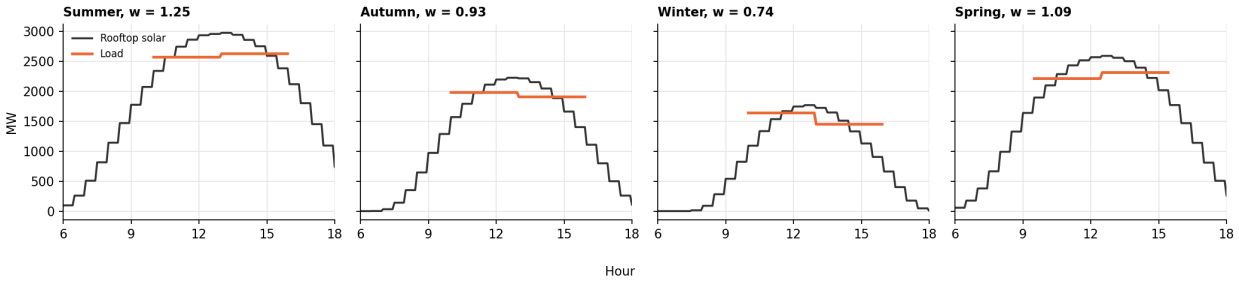


Figure 3: After the scaling: mean rooftop solar and the plan load $w_s L$.

3.2 Customer Mixes and Allocation

Load scale. To test other loads, the uptake is multiplied by a scale k: $k = 0.8, 1.0, 1.2$ give uptakes of 10.2%, 12.8% and 15.3%. Overload does not depend on k; the plan load grows in proportion (Figure 4).

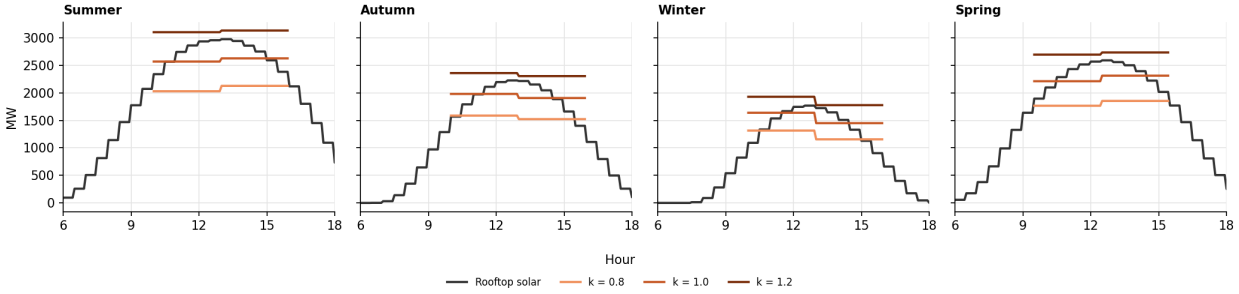


Figure 4: Mean rooftop solar and the plan load of two intervals at today's mix, for $k = 0.8$, 1.0 and 1.2.

3.2.1 Allocation strategy

An allocation strategy decides how each circuit splits its medium and large customers between the intervals. A split is the share of these customers given to each interval, for example 50% / 50%; each circuit rounds it to whole customers, separately for medium and large. We compare four kinds of strategy for each design:

- **Share rule:** every circuit uses the same split. Each share is a multiple of 5% (0%, 5%, …, 100%), which gives 21 possible rules for two intervals and 231 for three; all are evaluated.
- **Random:** every circuit composition draws its own split at random, uniformly over all splits; 300 such strategies per design.

- **Lowest overload:** every circuit composition gets its own split. If some split keeps the circuit under T, only those splits are used, otherwise the lowest-peak split; among these, the splits maximise O_{soak}.
- **Best soak:** every circuit composition gets its own split; the splits maximise O_{soak}, and among those O_{viol} is as low as possible.

The last two are found by coordinate ascent over compositions. The ideal point is fixed: no overload and $O_{\text{soak}} = 100\%$, all rooftop solar between 08:00 and 16:00 soaked with no excess load. The distance of a strategy from it is

$$\delta = \sqrt{O_{\text{viol}}^2 + \left(1 - O_{\text{soak}}\right)^2}, \tag{11}$$

with both as fractions. We show today's mix for the load scales $k = 0.8$, 1.0 and 1.2.

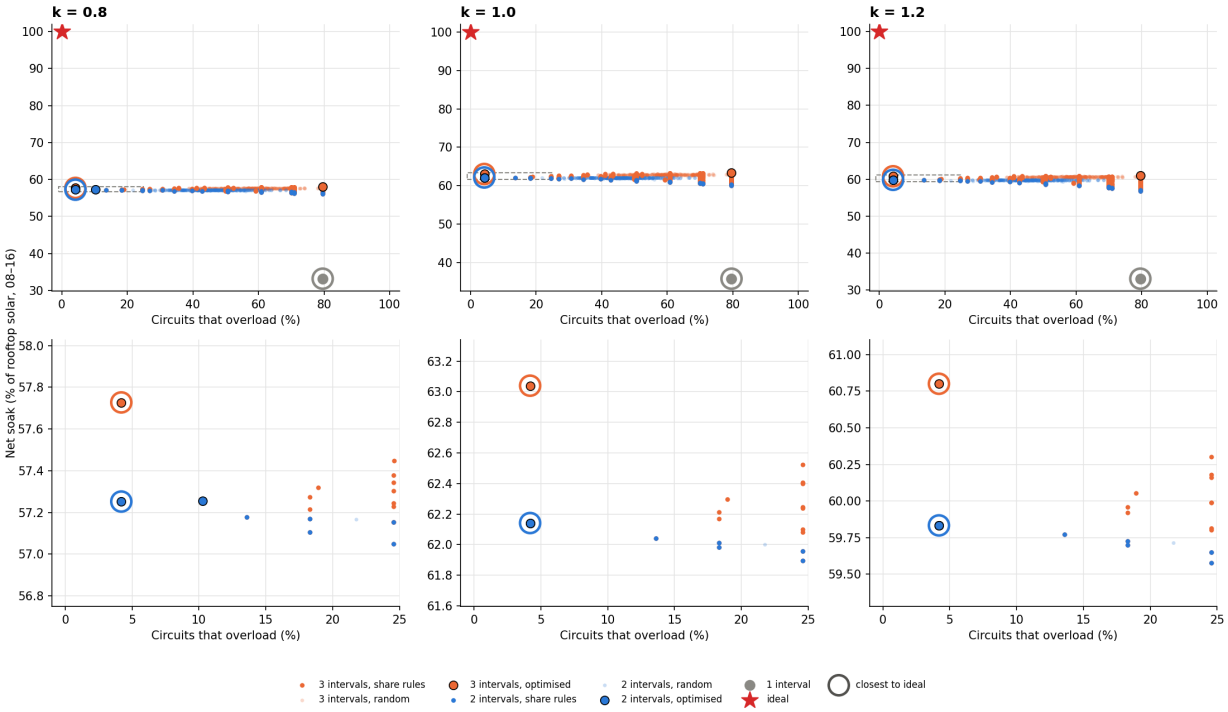


Figure 5: Overload and net soak of every strategy at today's mix, for each load scale. Grey: 1 interval; blue: 2 intervals; orange: 3 intervals. Red star: fixed ideal point; circles: the strategy closest to it for each design. Bottom: the dashed box enlarged.

				O_{soak} (%)			
Design	Strategy	Split (%)	O_{viol}	$k = 0.8$	$k = 1.0$	$k = 1.2$	δ, $k = 1.0$
1 interval	–	100	79.6%	33.2	35.8	33.0	1.02
2 intervals	share rule	53/47	13.6%	57.2	62.0	59.8	0.40
	random	60/40	21.8%	57.2	62.0	59.7	0.44
	lowest overload	50/50	4.2%	57.3	62.1	59.8	0.38
	best soak	50/50	4.2%*	57.3	62.1	59.8	0.38
3 intervals	share rule	53/0/47	13.6%	57.2	62.0	59.8	0.40
	random	38/26/36	36.0%	57.5	62.7	60.5	0.52
	lowest overload	23/47/30	4.2%	57.7	63.0	60.8	0.37
	best soak	10/76/14	79.6%	58.0	63.4	61.0	0.88

Best strategy of each kind (closest to the ideal point), today's mix. Split: share of medium and large customers in each interval at $k = 1.0$, averaged over circuits for the strategies that split each circuit differently. *10.3% at $k = 0.8$.

1. **Two or three intervals soak far more than one**: 62.1% against 35.8% of the rooftop solar at $k = 1.0$. One interval covers 3 hours and soaks only the solar in them; two and three intervals cover 6. Among the two- and three-interval strategies net soak differs by at most 1.4 points (Figure top row).
2. **Overload separates the designs too.** The lowest-overload strategy is closest to the ideal: 4.2% of circuits overload with two or three intervals, against 13.6% for the best share rule, 22–36% for the best random strategy and 79.6% for one interval.
3. **Three intervals use the middle interval.** The lowest-overload three-interval strategy puts 46–47% of medium and large customers in the middle and soaks 0.4–1.0 points more than two intervals at the same overload (bottom row).
4. **The allocation trade-off is not worth taking.** With three intervals the best-soak strategy moves 68–80% of medium and large customers into the middle, close to one interval, and overloads 79.6% of circuits for 0.2–0.4 points more net soak. With two intervals it overloads up to 10.3% of circuits for less than 0.01 points.
5. **Net soak is highest at the calibrated load ($k = 1.0$).** Below it the plan load leaves midday solar unsoaked; above it more of the load exceeds the rooftop output.

Where the middle interval helps. Figure 6 compares the plan load with rooftop solar in each season at today's mix and the calibrated uptake ($k = 1$). Three intervals cover the same hours as two, so the middle interval can only move load within them. The load of every customer at I_S is below the mean rooftop output in the middle of the day but above it in the morning and afternoon; moving customers into the middle interval shifts load to the sunniest hours, and three intervals soak more than two in every season, by 0.5 points of the season's 08:00–16:00 rooftop solar in summer, 1.7 in autumn, 2.7 in winter and 0.9 in spring. The best-soak split puts most customers in the middle (in winter all of them), which is the split that overloads circuits; an even split soaks less in every season (in winter 61.4% against 62.9%).

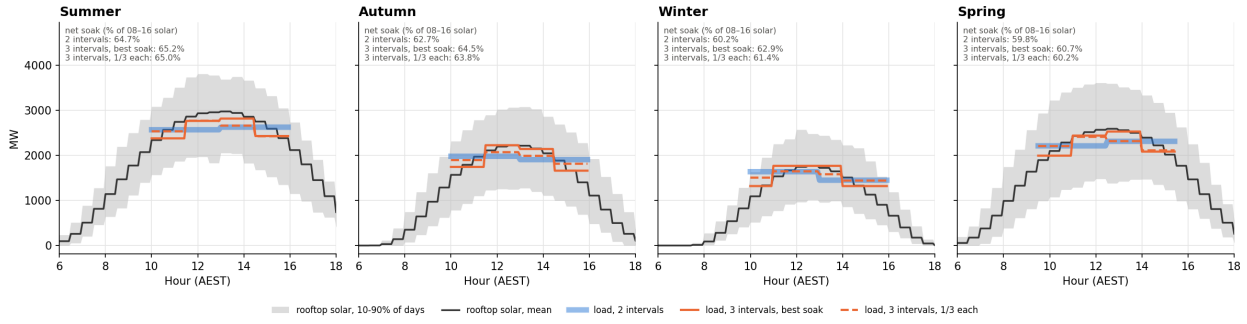


Figure 6: Rooftop solar (mean, and 10–90% of days) and the plan load counted for soak (scaled by w_s) at today's mix, per season. Blue: two intervals; orange: three intervals with the best-soak split (solid) and with customers split evenly (dashed; rounding to whole customers favours the first interval). Net soak per season, as % of that season's rooftop solar between 08:00 and 16:00, in the top left of each panel.

3.2.2 Customer mixes

Experiment. We vary the share of customers who add load in a free window, $\rho = p_M + p_L$, keeping medium and large customers in the ratio 30 : 1:

$$p_M = \tfrac{30}{31}\rho, \qquad p_L = \tfrac{1}{31}\rho, \qquad p_S = 1 - \rho, \qquad \rho \in \{0.31\, i/16 : i = 1, \ldots, 32\}, \tag{12}$$

so ρ runs from 1.9% to 62%; today's mix is $\rho = 31\%$. Each design traces one curve of $(O_{\text{viol}}, O_{\text{soak}})$.

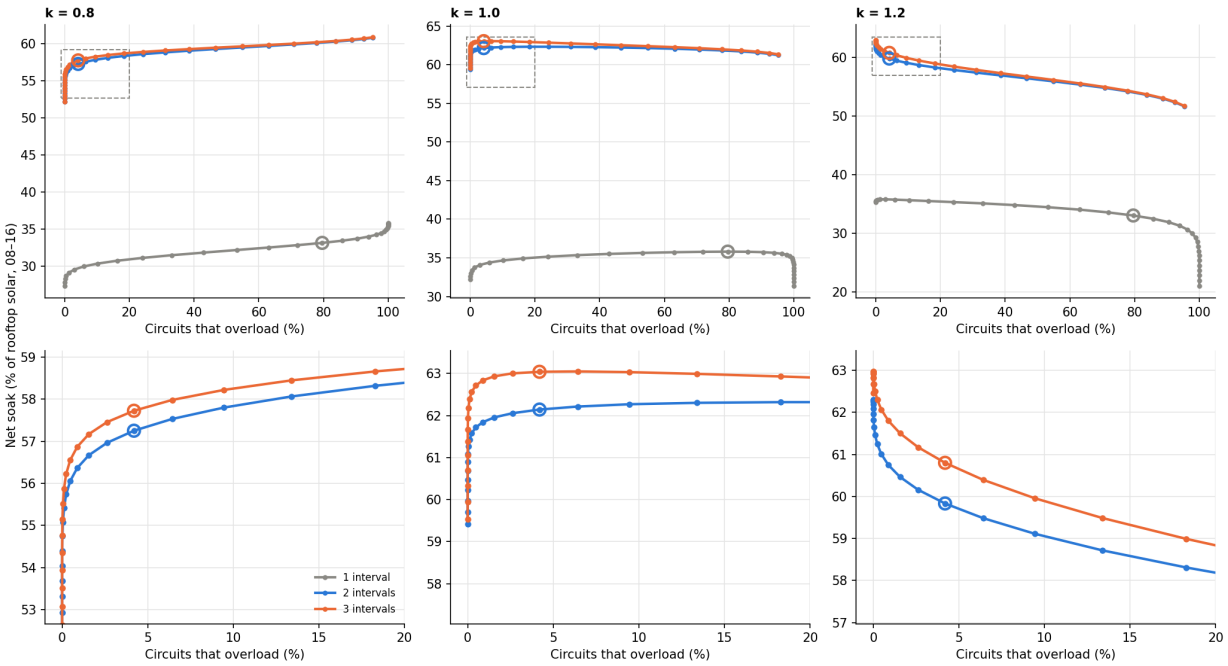


Figure 7: Overload against net soak as ρ grows, lowest-overload strategy, for each load scale. Circles: today's mix. Bottom: the dashed box enlarged.

	Today, $\rho = 31\%$			O_{soak} today (%)		
Design	O_{viol}	cap 1%	cap 5%	$k = 0.8$	$k = 1.0$	$k = 1.2$
1 interval	79.6%	7.8%	11.6%	33.2	35.8	33.0
2 intervals	4.2%	25.2%	31.0%	57.3	62.1	59.8
3 intervals	4.2%	25.2%	31.0%	57.7	63.0	60.8

Caps: largest ρ with overload at most 1% or 5%; lowest-overload strategy.

Trade-off.

1. **Allocation.** The best-soak strategy buys little soak with many overloads: with three intervals it moves most customers into the middle and overloads up to 79–81 points more circuits for at most 1.1–1.9% more net soak; with two intervals, up to 1–16 points more for at most 0.1%.
2. **One to two intervals.** At today's mix, overloaded circuits fall from 79.6% to 4.2% and net soak is 1.7–1.8 times higher. Under a 5% cap, ρ can rise from 12% to 31%.
3. **Net soak against ρ depends on the load scale.** For two and three intervals, at $k = 0.8$ the plan load stays below the solar and net soak rises up to $\rho = 62\%$ (61%); at $k = 1.0$ it peaks at $\rho = 33$–41% (62–63%), around today's mix; at $k = 1.2$ the plan load already exceeds the solar at low ρ and net soak falls as ρ grows. One interval stays at 27–36% at every ρ.
4. **Two to three intervals.** Overload is the same at every ρ, because the middle interval overlaps both others. Net soak rises by 0.4, 0.9 and 1.0 points at $k = 0.8$, 1.0 and 1.2 (bottom row), from moving load into the sunniest hours.

Probability of avoiding an upgrade. A circuit needs an upgrade if it overloads, so the probability that the scheme avoids one is $1 - O_{\text{viol}}$. We compute it for a share $x = p_L$ of heavy (large) customers, with $p_M = 30\%$ and $p_S = 1 - p_M - x$, and for N customers per line with limit

$T = 25\,\text{A} \times N$ (1000 A at $N = 40$), using the lowest-overload strategy, for one shared window, two assigned windows and three assigned windows (the middle one overlapping both).

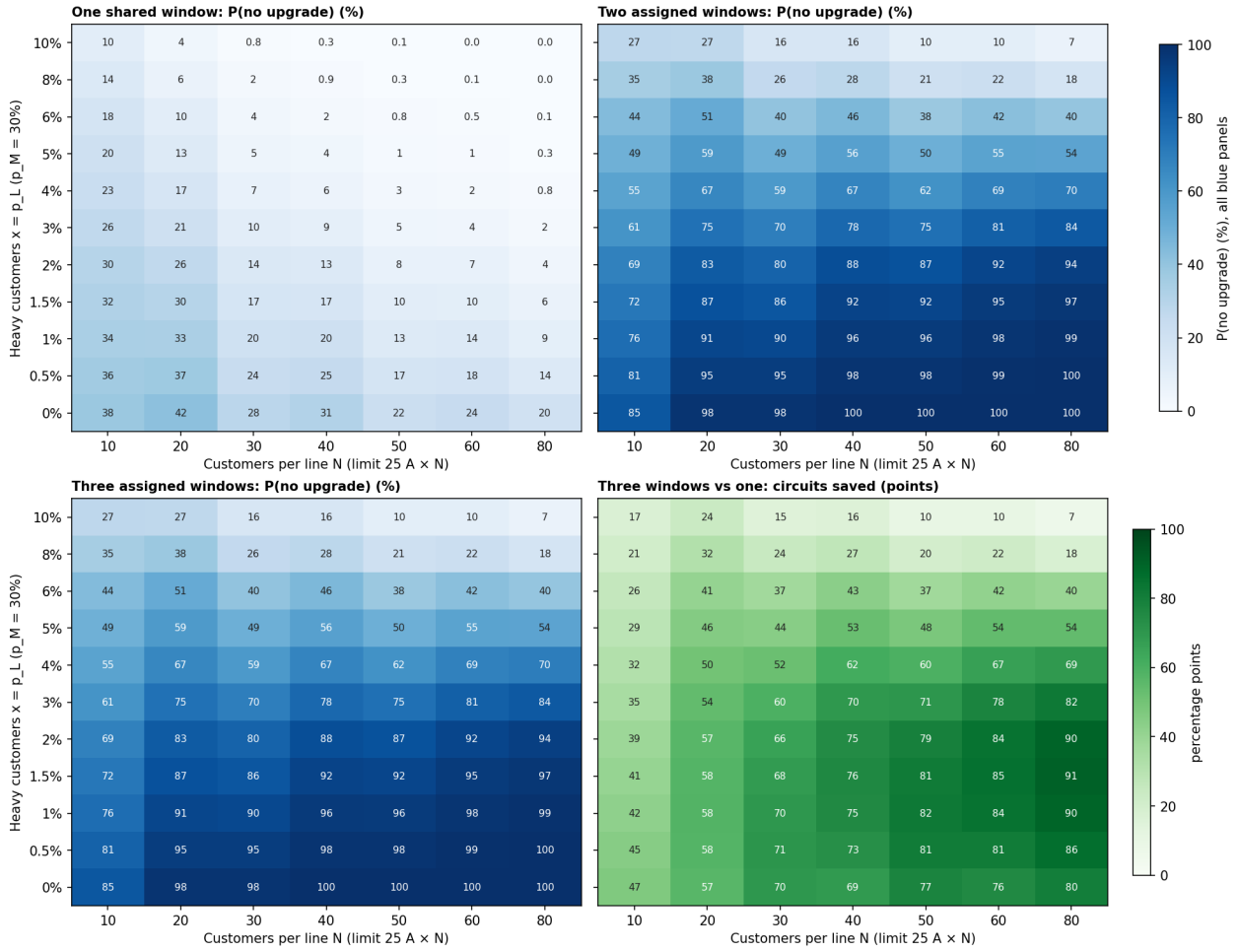


Figure 8: Probability of avoiding an upgrade, $1 - O_{\text{viol}}$ (%), against the share of heavy customers x and the customers per line N, for one, two and three windows. Green: three windows against one, in percentage points.

1. **Today's mix** ($x = 1\%$, $N = 40$): a line avoids an upgrade with probability 20% under one shared window and 96% under two assigned windows.
2. **Two windows keep the risk low while heavy customers are few.** The probability stays at or above 90% up to $x = 1.5\%$ for $N \geq 40$, and up to $x = 1\%$ for $N \geq 20$. One shared window stays below 42% everywhere, even with no heavy customers, because medium customers alone overload it.
3. **Three windows give exactly the same probability as two** in every cell, because the middle window overlaps both others and cannot lower a line's peak.
4. **Longer lines gain more.** At $x = 2\%$, assigned windows save 39 percentage points of lines at $N = 10$ and 90 at $N = 80$.
5. **From** $x = 6\%$ more than half of lines need an upgrade under either scheme (all N except $N = 20$).

The pattern is not smooth in N because the limit and the currents only allow whole numbers of customers.